Machine Learning for social sciences

Introduction to Machine Learning

William Aboucaya

Why are we here?

Who are you?

- Who has prior experience with coding?

Who are you?

- Who has prior experience with coding?
- Who has prior experience with data science? machine learning?

Introduction

Who are you?

- Who has prior experience with coding?
- Who has prior experience with data science? machine learning?

If you have prior experience with Python, this catch-up class is most likely not for you. Here, we will see the basics of Python and how to use it properly in a data science project.

Who are you?

- Who has prior experience with coding?
- Who has prior experience with data science? machine learning?

If you have prior experience with Python, this catch-up class is most likely not for you. Here, we will see the basics of Python and how to use it properly in a data science project.
But first, we'll quickly motivate why we're going to learn a programming language: data science and machine learning

What is machine learning?

What is machine learning?

- A part of the field of artificial intelligence?

What is machine learning?

- A part of the field of artificial intelligence?
- A field of methods aiming at making sense of large data?

What is machine learning?

- A part of the field of artificial intelligence?
- A field of methods aiming at making sense of large data?
- Extrapolation on new data based on previous trends?

What is machine learning?

- A part of the field of artificial intelligence?
- A field of methods aiming at making sense of large data?
- Extrapolation on new data based on previous trends?

All of it and more!

Intuitive definition:

Machine learning is the study and practice of algorithms that *automatically* extract patterns from data and improve their performance on tasks through experience.

Examples:

- Supervised: image classification, house price regression
- Unsupervised: clustering, dimensionality reduction
- Reinforcement: game playing, robotics, LLM alignment

Formal (concise) definition:

An algorithm is said to *learn* from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.

Key ingredients

- **Data / Experience (E):** labeled or unlabeled examples, interactions, sensor readings
- **Task (T):** classification, regression, clustering, control, etc.
- **Performance (P):** accuracy, MSE, reward, F1 score, etc.
- **Model / Hypothesis:** $f_\theta(x)$ with parameters θ learned from data

Outline of the catch-up session

1. Introduction $\Leftarrow$ **You are here**
2. Hands-on Python tutorial
3. Working on a Python data science / machine learning project

Why do we train models?

Two Paradigms in AI

Expert Systems (Symbolic AI)

Knowledge is **explicitly encoded** as rules provided by human experts.

Trained Models (Statistical / Machine Learning)

Knowledge is **implicitly learned** from data by optimizing model parameters

Two Paradigms in AI

Expert Systems (Symbolic AI)

Knowledge is **explicitly encoded** as rules provided by human experts.

Trained Models (Statistical / Machine Learning)

Knowledge is **implicitly learned** from data by optimizing model parameters

- Expert systems: “top-down” reasoning
- Trained models: “bottom-up” learning

Expert Systems

Core idea: emulate human decision-making via a rule base and inference engine

Knowledge representation: IF–THEN rules

Advantages:

- Transparent reasoning
- Easy to audit and modify

Limitations:

- Hard to scale and maintain
- Fragile in noisy or uncertain domains

Trained Models (Machine Learning)

Core idea: learn patterns from large datasets

Examples: neural networks, decision trees, transformers

Knowledge representation: parameters and weights

Advantages:

- Adapt well to complex and unstructured data
- Automatically improve with more data

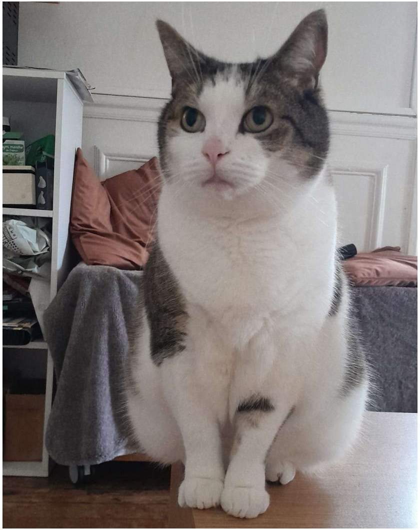
Limitations:

- Lack interpretability ("black box")
- Require large amounts of data and computing power

Example: Identifying cat pictures

Using an expert system:

if *[Entity] number_of_limbs 4*
    and
  [Entity] has [Fur]
    and
  [Entity] has [Whiskers] **then**
   [Entity] is [Cat]
else
   [Entity] is_not [Cat]



Example: Identifying cat pictures

Using an expert system:

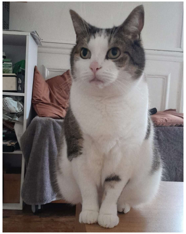
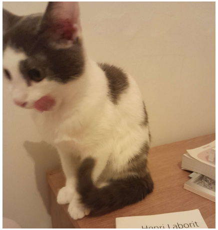
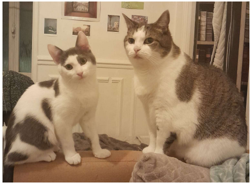
if *[Entity] number_of_limbs 4*
  and
 [Entity] has [Fur]
  and
 [Entity] has [Whiskers] **then**
  [Entity] is [Cat]
else
  [Entity] is_not [Cat]

But outliers can be hard to anticipate!

Example: Identifying cat pictures

Using machine learning:

Cats

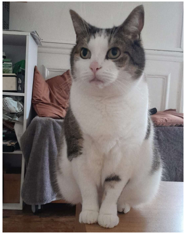

Non-cats

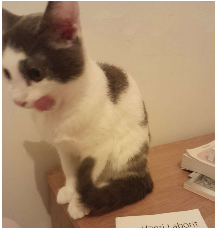


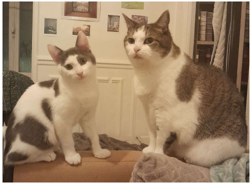

Example: Identifying cat pictures

Using machine learning:

Cats







Non-cats

Generalize from hundreds of pictures of cats and non-cats!

Summary

Expert Systems

- Rule-based
- Transparent
- Brittle

Trained Models

- Data-driven
- Opaque
- Robust

Summary

Expert Systems

- Rule-based
- Transparent
- Brittle

Trained Models

- Data-driven
- Opaque
- Robust

Modern Trend: Neuro-Symbolic AI

Combining explicit reasoning (rules, logic, knowledge bases) with learned representations (trained neural networks) to get the best of both worlds.

What we need for the following

Software requirements:

1. Python 3.12
2. A development environment (here, we will use Visual Studio Code)
3. UV, a Python environment manager

Let's install all these!